

Alessandro Contri

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Summary

PhD candidate in Computational Mathematics with 4+ years of experience developing physics-based models and numerical methods for coupled multiphysics systems. Specialized in finite element and particle-based methods for realistic geometries and evolving domains. Trained in building high-performance scientific software in Python and C++ (e.g., NgSolve, Kratos Multiphysics) with a focus on reproducibility and a proven record of cross-disciplinary collaboration.

Education

Norwegian University of Science and Technology (NTNU)	Norway
PhD Candidate - Computational Mathematics. Supervisors: André Massing, Elena Celledoni	Jan 2022 - Present
University of California San Diego (UCSD)	USA
Visiting Scholar. Supervisor: Padmini Rangamani	Jan 2024 - Jun 2024
University of Padova	Italy
Master of Science - Mathematical Engineering. Supervisors: Antonia Larese, Stefano Lanzoni	Sep 2018 - Dec 2021
University of Padova	Italy
Bachelor of Science - Aerospace Engineering. Supervisor: Stefano Casotto	Sep 2015 - Sep 2018

Experience

PhD Research	Jan 2022 - Present
- Developed a FEM framework for simulating cell reshaping phenomena using microscopy-derived geometries. Integrated mixed-dimensional and gradient flow solvers to support complex multiphysics workflows. - Built Python package within the high-performance NgSolve environment (~500 users), with emphasis on reproducibility, documentation, and simulation pipeline design. - Published new error estimates for time-stepping schemes (BDF1–6) applied to the transient Stokes equations in SIAM Numerical Analysis. - Collaborated with biophysics researchers at UC San Diego and NTNU to identify key biological processes and derive appropriate PDE models for the framework. - Supervised and assisted student cohorts in numerical courses (up to 1000 students).	
Master Thesis Research	Jan 2021 - Dec 2021
- Designed and implemented a new constitutive model for fluid-like materials in the Material Point Method, enabling simulation of large-deformation fluid–structure interaction (FSI) scenarios such as avalanches and debris flows. - Integrated fluid dynamics modules into Kratos Multiphysics (~1000 users), ensuring compatibility with solid mechanics solvers and stability under incompressibility constraints (Link to repository).	
Bachelor Thesis Research	Jan 2018 - Sep 2018
- Reconstructed interplanetary spacecraft trajectories (Cassini, Rosetta) using NASA SPICE kernels. - Compared influence of multiple physical factors (radiation pressure, general relativity) on orbit design.	

Publications

- Alessandro Contri, André Massing, and Padmini Rangamani. Mechanochemical Feedback between Cell Shape and Intracellular Mechanics Revealed by a Finite-Element Framework, 2026 - **Submitted**
- Alessandro Contri, André Massing, and Padmini Rangamani. A finite element framework for solving coupled multiphysics problems with moving boundaries in cell biophysics. *Computer Methods in Applied Mechanics and Engineering*, 459:119071, September 2026. [\[Link to article\]](#) - **Published**
- Alessandro Contri, Balázs Kovács, and André Massing. Error Analysis of BDF 1–6 Time-Stepping Methods for the Transient Stokes Problem: Velocity and Pressure Estimates. *SIAM Journal on Numerical Analysis*, 63(4):1586–1616, August 2025. [\[Link to article\]](#) - **Published**
- Alessandro Contri, Christina Taylor, Justin Tso, and Ingeborg Gjerde. *Inducing Flow Instabilities in Aneurysm Geometries via the Reynolds-Orr Method*, pages 79–89. Springer Nature Switzerland, Cham, 2023. [\[Link to report\]](#) - **Published**.

Contributed presentations

- Contri, A.**, Massing, A., Rangamani, P., 2026. A finite element framework for bulk-surface coupled PDEs to solve moving boundary problems in biophysics. Oral presentation at A Roadmap towards Developing Mechanobiochemical Models for Single and Collective Cell Migration through Complex Non-Isotropic Environments. Banff, Canada.

- **Contri, A.**, Massing, A., Rangamani, P., 2024. A biologically-driven FEM framework for coupled PDEs on realistic geometries. Oral presentation at 30th Biennial Numerical Analysis Conference. Glasgow, UK.
- **Contri, A.**, Massing, A., Rangamani, P., 2024. Towards bulk-surface evolving FEM for the simulation of dendritic spines growth on realistic geometries. Invited Talk at Finite Elements for Cell and Tissue Morphogenesis. La Villa Clythia, Fréjus, France.
- **Contri, A.**, Kovács, B., Massing, A., 2024. Error analysis of BDF 1-6 time-stepping methods for the transient Stokes problem: velocity and pressure estimates. Poster Session at PDEs and Geometry: Numerical Aspects, ICERM, Boston, USA.
- **Contri, A.**, Kovács, B., Massing, A., 2023. Error analysis of BDF 1-6 time-stepping methods for the transient Stokes problem: velocity and pressure estimates. Oral presentation at MAGIC - Manifolds and Geometric Integration Colloquia, Ilsetra, Norway.

Outreach events participation

- *26w5573 - A Roadmap towards Developing Mechanobiochemical Models for Single and Collective Cell Migration through Complex Non-Isotropic Environments.* Banff, Canada. 10-15/5/26.
- *30th Biennial Numerical Analysis Conference.* Glasgow, UK. 24-27/6/25.
- *Finite Elements for Cell and Tissue Morphogenesis.* La Villa Clythia, Fréjus, France. 9-13/9/24.
- *PDEs and Geometry: Numerical Aspects.* Boston. USA. 11-15/3/24.
- *International Workshop Vector- and Tensor-Valued Surface PDEs.* Dresden, Germany. 29/11/23-01/12/23.
- *MAGIC - Manifolds and Geometric Integration Colloquia.* Ilsetra, Norway. 27/02/23-03/03/23.
- *SIAM Hackathon (Part of CSE23).* Amsterdam, Netherlands. 25-26/02/23.
- *ESGI - European Study Group with Industry.* Ålesund, Norway. 13-17/06/22.
- *ECCOMAS 2022 - European Community on Computational Methods in Applied Science.* Oslo, Norway. 05-09/06/22.

Awards

- Travel & Accommodation Award, ICERM, Brown University. 11-15/3/24.
- Travel & Accommodation Award, Centre National de la Recherche Scientifique. 9-13/9/24.

Teaching Assistantship

- *TMA4220 - Numerical Solution of Partial Differential Equations Using Element Methods.* Autumn 2024.
- *TMA4212 - Numerical Solution of Differential Equations by Difference Methods.* Spring 2022 & 2023.
- *TMA4130/TMA4135 - Mathematics 4N/4D - Differential equations and Fourier analysis.* Fall 2022 & 2023.
- *TMA4125 - Mathematics 4N - Differential equations and Fourier analysis.* Spring 2023.

Technical skills

- Numerical Modeling: Finite Element Method (FEM), Material Point Method (MPM), PDE solvers, Multiphysics coupling
- Complementary Tools: Stochastic processes, Non-equilibrium statistical mechanics, Deep Learning
- Programming & Tools: Python, C++, NgSolve, Kratos Multiphysics, NumPy, SciPy, Git, Parallel Programming
- Applications: Fluid Dynamics, Solid Mechanics, Fluid-Structure Interaction, Mixed-Dimensional PDEs
- Software Engineering: Python packaging, Scientific code development, Reproducibility, Documentation

Languages

- Italian (mothertongue), English (C1) and -litt- Norwegian (A2)